Experimento 007 - Geracao guiada com novidade C2DB e relaxacao M3GNet

Objetivo

Gerar candidatos UWBG por substituicao em prototipos C2DB, manter materiais conhecidos como controles de validacao e separar a tabela de descoberta em composicoes realmente novas. A rodada atual tambem relaxa candidatos selecionados com M3GNet antes de recalcular o gap com o MEGNet treinado.

Implementacao atual

- O indice `outputs/c2db_material_index.csv` compara formula reduzida e layergroup contra o C2DB.
- `novelty_class` separa `known_material`, `known_composition_new_layergroup` e `new_composition`.
- `top_novel_candidates.csv` contem somente `new_composition`.
- Candidatos conhecidos permanecem em `gap_bin_samples_all.csv` para validacao/calibracao.
- O MEGNet de gap usa o `matgl` do conda `matgl-tcc`, isto e, o mesmo ambiente do treino.
- A relaxacao M3GNet roda em subprocesso via `08_guided_generation/relax_m3gnet_batch.py`, usando `final/vendor/matgl_src` e o modelo local `final/models/M3GNet-PES-MatPES-PBE-2025.2`.
- `relax_cell=False` preserva a celula/vacuo 2D; a relaxacao altera posicoes atomicas.

Resultados gerais

- Candidatos baseline: 1083; taxa UWBG predita: 61.7%.
- Candidatos guiados: 697; taxa UWBG predita: 78.9%.
- Guiados `new_composition`: 323.
- Guiados `known_material`: 297.
- Guiados `known_composition_new_layergroup`: 77.
- Top candidatos finais: 50, todos `new_composition`.
- Distribuicao guiada por faixa de gap nao relaxado: {'0-2': 34, '10-12': 7, '2-4': 225, '4-6': 320, '6-8': 93, '8-10': 15}.
- Relaxacao M3GNet: {'relaxed': 90}.
- Deltas pos-relaxacao: media -0.801 eV, mediana -0.600 eV, minimo -5.572 eV, maximo 3.345 eV.
- Casos relaxados ainda UWBG (`gap_pred_relaxed >= 3.4 eV`): 74/90.
- Casos novos relaxados ainda UWBG: 50/61.
- Distribuicao dos gaps relaxados: {'0-2': 5, '2-4': 22, '4-6': 26, '6-8': 19, '8-10': 14, '10-12': 4, '>12': 0}.

Interpretacao da relaxacao

A relaxacao reduziu o gap predito na media, mas nao de forma uniforme. Isso confirma que o ranking baseado apenas no prototipo substituido pode superestimar candidatos, principalmente quando `substitution_risk=high`. Ainda assim, 50 de 61 composicoes novas relaxadas permaneceram acima de 3.4 eV, entao a geracao continua produzindo um conjunto util de candidatos UWBG, agora com triagem mais conservadora.

Top composicoes novas por gap relaxado

formula	proto_formula	proto_layergroup	substitution	gap_pred_unrelaxed	gap_pred_relaxed	gap_pred_relax_delta	s_chem_rf	m3gnet_e
BeB2F8	MgB2F8	p-3m1	Mg→Be	10.687	10.858	0.171	0.830	-5.599
MgAl2F8	MgB2F8	p-3m1	B→Al	9.775	10.631	0.856	0.865	-5.269
BF3	AlF3	p-31m	Al→B	11.405	10.122	-1.283	0.580	-5.750
BaB2F8	MgB2F8	p-3m1	Mg→Ba	10.039	9.750	-0.289	0.885	-5.824
BOF	GaOF	pmmn	Ga→B	8.913	9.502	0.590	0.505	-6.917
NaSrF3	NaSrI3	p-31m	I→F	9.174	9.431	0.257	0.930	-4.579
LiMgF3	LiMgBr3	p-31m	Br→F	9.694	9.392	-0.302	0.955	-4.555
CaB2F8	MgB2F8	p-3m1	Mg→Ca	10.521	9.372	-1.149	0.935	-5.682
NaCaF3	NaCaI3	p-31m	I→F	9.788	9.054	-0.734	0.915	-4.667
ZrZnF6	HfZnF6	p312	Hf→Zr	8.253	7.728	-0.525	0.685	-5.217
BClO	GaClO	pmmn	Ga→B	7.033	7.727	0.694	0.515	-6.052
SrB2F8	MgB2F8	p-3m1	Mg→Sr	10.266	7.668	-2.598	0.925	-5.618
Sr(BH4)2	Mg(BH4)2	p-3m1	Mg→Sr	7.190	7.248	0.058	0.565	-4.025
NaCaCl3	NaCaI3	p-31m	I→Cl	7.668	7.065	-0.603	0.825	-3.546
Ca(BH4)2	Mg(BH4)2	p-3m1	Mg→Ca	7.170	6.978	-0.191	0.510	-4.072
LiCaCl3	LiMgCl3	p-31m	Mg→Ca	7.208	6.875	-0.334	0.855	-3.543
Be(BH4)2	Mg(BH4)2	p-3m1	Mg→Be	8.437	6.731	-1.705	0.615	-4.071
BeClF	CaClF	p4/nmm	Ca→Be	7.756	6.610	-1.146	0.840	-4.533

formula	proto_formula	proto_layergroup	substitution	gap_pred_unrelaxed	gap_pred_relaxed	gap_pred_relax_delta	s_chem_rf	m3gnet_e
Ba(BH4)2	Mg(BH4)2	p-3m1	Mg→Ba	6.730	6.602	-0.127	0.575	-4.181
LiBeCl3	LiMgCl3	p-31m	Mg→Be	7.610	6.422	-1.188	0.960	-3.589

Comparacao DFT externa

LiF e BaF2 continuam apenas como comparacao externa; eles nao entram no treino porque sao dois pontos e ambos diagnosticam problemas de geracao/ranking, nao uma distribuicao de treino. A comparacao abaixo mostra que a maior parte da superestimacao vinha da geometria nao relaxada.

formula	proto_uid	proto_layergroup	gap_pred_original	gap_pred_relaxed	gap_dft	error_original_pred_minus_dft	error_relaxed_pred_minus_
LiF	6BrLi-2	p3m1	9.624	8.414	8.460	1.164	-0.046
BaF2	1BaBr2-2	p-6m2	8.177	7.431	7.530	0.647	-0.099

Tabelas por faixa de gap

Arquivos completos: `gap_bin_samples_all.csv`, `gap_bin_samples_new_composition.csv` e `relaxation_results.csv`.

Amostras guiadas - todos os candidatos

0-2 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_de
0-2	1	Hf3ScSb3Br4O	Hf3ScBr4N3O	cm11	N→Sb	1.994	1.530	-0.463
0-2	2	LiAl(S3N)2	LiAl(PS3)2	c211	P→N	1.975	2.142	0.167
0-2	3	TiNiF6	HfNiF6	p312	Hf→Ti	1.936	5.281	3.345
0-2	4	CsCSN	NaCSN	pm2_1n	Na→Cs	1.930	3.861	1.931
0-2	5	YI3	YCl3	p-31m	Cl→I	1.856	3.155	1.299
0-2	6	RbMnBr3	LiMnBr3	p-31m	Li→Rb	1.837	0.916	-0.920
0-2	7	Hf3ScP3Br4O	Hf3ScBr4N3O	cm11	N→P	1.829	2.392	0.564
0-2	8	SrVIBr3	SrVICl3	pm11	Cl→Br	1.815	2.527	0.712
0-2	9	CsF5	CsIF4	p4/mbm	I→F	1.813	3.913	2.100
0-2	10	TiPdF6	HfPdF6	p312	Hf→Ti	1.755	3.157	1.402

2-4 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_del
2-4	1	Ca2Mg(SBr)2	SrCa2(SBr)2	p-3m1	Sr→Mg	3.998	4.490	0.492
2-4	2	YCuBr4	YCuCl4	p-1	Cl→Br	3.997	3.695	-0.302
2-4	3	ZrPdF6	HfPdF6	p312	Hf→Zr	3.995	2.511	-1.484
2-4	4	BaHCl	CaHCl	p4/nmm	Ca→Ba	3.992	3.951	-0.041
2-4	5	BeP2(SCl)4	BeP2(SF)4	c211	F→Cl	3.988	3.682	-0.306
2-4	6	VClF	VBrCl	p2_1/m11	Br→F	3.979	2.322	-1.656
2-4	7	HfNF	ZrNF	p-3m1	Zr→Hf	3.976	1.266	-2.710
2-4	8	BiPO3	BiSbO3	p11a	Sb→P	3.976	4.228	0.251
2-4	9	SnNClO3	SnPClO3	p11a	P→N	3.964	3.739	-0.225
2-4	10	CdHOF	CdHClO	p3m1	Cl→F	3.957	4.482	0.525

4-6 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
4-6	1	NaCSN	KCSN	pm2_1n	K→Na	5.997	3.898	-2.099
4-6	2	AsPO4	SbPO4	p2_1/m11	Sb→As	5.996	3.907	-2.088
4-6	3	NaClO2	LiClO2	p-42_1m	Li→Na	5.987	4.554	-1.433
4-6	4	LiBrO2	LiClO2	p-42_1m	Cl→Br	5.973	0.619	-5.354
4-6	5	NaHO	LiHO	p4/nmm	Li→Na	5.963	4.888	-1.075

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
4-6	6	ScCl3	ScBr2Cl	cm11	Br→Cl	5.960	4.546	-1.414
4-6	7	Ga2Cl5F	Ga2BrCl5	cm11	Br→F	5.952	4.225	-1.727
4-6	8	CsCl	CsBr	p4/nmm	Br→Cl	5.951	4.881	-1.070
4-6	9	ZrO2	TiO2	p-3m1	Ti→Zr	5.949	6.158	0.209
4-6	10	ScBrO	ScOF	pmmn	F→Br	5.947	4.886	-1.060

6-8 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
6-8	1	B2O3	Ga2O3	pm2_1n	Ga→B	7.947	8.096	0.148
6-8	2	NaF	NaI	p4/mmm	I→F	7.916	7.594	-0.322
6-8	3	BeClF	CaClF	p4/nmm	Ca→Be	7.756	6.610	-1.146
6-8	4	NaCaCl3	NaCaI3	p-31m	I→Cl	7.668	7.065	-0.603
6-8	5	LiBeCl3	LiMgCl3	p-31m	Mg→Be	7.610	6.422	-1.188
6-8	6	NaSrCl3	NaSrI3	p-31m	I→Cl	7.578	6.126	-1.452
6-8	7	HfNiF6	HfNiCl6	p312	Cl→F	7.556	4.874	-2.682
6-8	8	BeCl2	MgCl2	p-3m1	Mg→Be	7.521	6.320	-1.201
6-8	9	ScClF2	ScBr2Cl	cm11	Br→F	7.470	5.282	-2.188
6-8	10	MnF2	MnCl2	p-3m1	Cl→F	7.379	4.361	-3.019

8-10 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
8-10	1	YF3	YI3	p-31m	I→F	9.883	9.587	-0.296
8-10	2	NaCaF3	NaCaI3	p-31m	I→F	9.788	9.054	-0.734
8-10	3	MgAl2F8	MgB2F8	p-3m1	B→Al	9.775	10.631	0.856
8-10	4	LiMgF3	LiMgBr3	p-31m	Br→F	9.694	9.392	-0.302
8-10	5	CaF2	CaBr2	pmmn	Br→F	9.682	8.741	-0.941
8-10	6	LiF	LiBr	p3m1	Br→F	9.624	8.414	-1.209
8-10	7	MgF2	MgCl2	p-3m1	Cl→F	9.360	8.982	-0.378
8-10	8	ScF3	ScCl3	p6/mmm	Cl→F	9.353	9.181	-0.173
8-10	9	NaSrF3	NaSrI3	p-31m	I→F	9.174	9.431	0.257
8-10	10	SrF2	SrBr2	pmmn	Br→F	9.163	8.435	-0.727

10-12 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
10-12	1	AlF3	AlCl3	p-62m	Cl→F	11.519	10.422	-1.097
10-12	2	BF3	AlF3	p-31m	Al→B	11.405	10.122	-1.283
10-12	3	BeF2	MgF2	p-3m1	Mg→Be	11.064	9.961	-1.103
10-12	4	BeB2F8	MgB2F8	p-3m1	Mg→Be	10.687	10.858	0.171
10-12	5	CaB2F8	MgB2F8	p-3m1	Mg→Ca	10.521	9.372	-1.149
10-12	6	SrB2F8	MgB2F8	p-3m1	Mg→Sr	10.266	7.668	-2.598
10-12	7	BaB2F8	MgB2F8	p-3m1	Mg→Ba	10.039	9.750	-0.289

>12 eV

Nenhum caso nesta faixa.

Amostras guiadas - apenas new_composition

0-2 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_de
0-2	1	Hf3ScSb3Br4O	Hf3ScBr4N3O	cm11	N→Sb	1.994	1.530	-0.463
0-2	2	LiAl(S3N)2	LiAl(PS3)2	c211	P→N	1.975	2.142	0.167
0-2	3	TiNiF6	HfNiF6	p312	Hf→Ti	1.936	5.281	3.345
0-2	4	CsCSN	NaCSN	pm2_1n	Na→Cs	1.930	3.861	1.931
0-2	5	RbMnBr3	LiMnBr3	p-31m	Li→Rb	1.837	0.916	-0.920
0-2	6	Hf3ScP3Br4O	Hf3ScBr4N3O	cm11	N→P	1.829	2.392	0.564
0-2	7	SrVIBr3	SrVICl3	pm11	Cl→Br	1.815	2.527	0.712
0-2	8	CsF5	CsIF4	p4/mbm	I→F	1.813	3.913	2.100
0-2	9	TiPdF6	HfPdF6	p312	Hf→Ti	1.755	3.157	1.402
0-2	10	MgVICl3	SrVICl3	pm11	Sr→Mg	1.683	2.213	0.531

2-4 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_del
2-4	1	Ca2Mg(SBr)2	SrCa2(SBr)2	p-3m1	Sr→Mg	3.998	4.490	0.492
2-4	2	YCuBr4	YCuCl4	p-1	Cl→Br	3.997	3.695	-0.302
2-4	3	BeP2(SCl)4	BeP2(SF)4	c211	F→Cl	3.988	3.682	-0.306
2-4	4	SnNClO3	SnPClO3	p11a	P→N	3.964	3.739	-0.225
2-4	5	SrCa2(SI)2	SrCa2(SBr)2	p-3m1	Br→I	3.951	3.838	-0.113
2-4	6	BaLiI3	LiMgI3	p-31m	Mg→Ba	3.945	3.482	-0.463
2-4	7	SrVCl4	SrVICl3	pm11	I→Cl	3.943	2.964	-0.980
2-4	8	Sr3(SBr)2	SrCa2(SBr)2	p-3m1	Ca→Sr	3.923	4.331	0.409
2-4	9	BeAgBr3	CaAgBr3	p-31m	Ca→Be	3.914	4.184	0.270
2-4	10	BaCuBr3	MgCuBr3	p-31m	Mg→Ba	3.914	3.043	-0.871

4-6 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
4-6	1	Ga2Cl5F	Ga2BrCl5	cm11	Br→F	5.952	4.225	-1.727
4-6	2	MgAgF3	MgAgBr3	p-31m	Br→F	5.943	0.371	-5.572
4-6	3	BeHCl	BaHCl	p4/nmm	Ba→Be	5.936	5.342	-0.595
4-6	4	RbClO2	LiClO2	p-42_1m	Li→Rb	5.933	5.380	-0.553
4-6	5	CsGeHO3	CsHCO3	p2_111	C→Ge	5.924	4.637	-1.286
4-6	6	BeIBr	BaIBr	p3m1	Ba→Be	5.914	4.700	-1.214
4-6	7	BaHF	BaHCl	p4/nmm	Cl→F	5.901	6.231	0.330
4-6	8	Hf3O5F2	Hf3Br2O5	p1	Br→F	5.894	6.136	0.242
4-6	9	MgHCl	BaHCl	p4/nmm	Ba→Mg	5.862	6.254	0.392
4-6	10	LiBeBr3	LiMgBr3	p-31m	Mg→Be	5.857	4.276	-1.581

6-8 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
6-8	1	BeClF	CaClF	p4/nmm	Ca→Be	7.756	6.610	-1.146
6-8	2	NaCaCl3	NaCaI3	p-31m	I→Cl	7.668	7.065	-0.603
6-8	3	LiBeCl3	LiMgCl3	p-31m	Mg→Be	7.610	6.422	-1.188
6-8	4	NaSrCl3	NaSrI3	p-31m	I→Cl	7.578	6.126	-1.452
6-8	5	ScClF2	ScBr2Cl	cm11	Br→F	7.470	5.282	-2.188
6-8	6	SiPO3F	GePO3F	p11a	Ge→Si	7.292	3.241	-4.051
6-8	7	LiCaCl3	LiMgCl3	p-31m	Mg→Ca	7.208	6.875	-0.334

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
6-8	8	Sr(BH4)2	Mg(BH4)2	p-3m1	Mg→Sr	7.190	7.248	0.058
6-8	9	Ca(BH4)2	Mg(BH4)2	p-3m1	Mg→Ca	7.170	6.978	-0.191
6-8	10	PCO3F	GePO3F	p11a	Ge→C	7.165	4.227	-2.938

8-10 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
8-10	1	NaCaF3	NaCaI3	p-31m	I→F	9.788	9.054	-0.734
8-10	2	MgAl2F8	MgB2F8	p-3m1	B→Al	9.775	10.631	0.856
8-10	3	LiMgF3	LiMgBr3	p-31m	Br→F	9.694	9.392	-0.302
8-10	4	NaSrF3	NaSrI3	p-31m	I→F	9.174	9.431	0.257
8-10	5	BOF	GaOF	pmmn	Ga→B	8.913	9.502	0.590
8-10	6	Be(BH4)2	Mg(BH4)2	p-3m1	Mg→Be	8.437	6.731	-1.705
8-10	7	ZrZnF6	HfZnF6	p312	Hf→Zr	8.253	7.728	-0.525

10-12 eV

gap_bin	gap_bin_sample_rank	formula	proto_formula	proto_layergroup	substitution	gap_pred	gap_pred_relaxed	gap_pred_relax_delta
10-12	1	BF3	AlF3	p-31m	Al→B	11.405	10.122	-1.283
10-12	2	BeB2F8	MgB2F8	p-3m1	Mg→Be	10.687	10.858	0.171
10-12	3	CaB2F8	MgB2F8	p-3m1	Mg→Ca	10.521	9.372	-1.149
10-12	4	SrB2F8	MgB2F8	p-3m1	Mg→Sr	10.266	7.668	-2.598
10-12	5	BaB2F8	MgB2F8	p-3m1	Mg→Ba	10.039	9.750	-0.289

>12 eV

Nenhum caso nesta faixa.

Top candidatos novos nao relaxados

formula	proto_formula	proto_layergroup	substitution	gap_pred	s_chem_rf	substitution_risk	novelty_class
BF3	AlF3	p-31m	Al→B	11.405	0.580	medium	new_composition
BeB2F8	MgB2F8	p-3m1	Mg→Be	10.687	0.830	medium	new_composition
CaB2F8	MgB2F8	p-3m1	Mg→Ca	10.521	0.935	high	new_composition
SrB2F8	MgB2F8	p-3m1	Mg→Sr	10.266	0.925	high	new_composition
BaB2F8	MgB2F8	p-3m1	Mg→Ba	10.039	0.885	high	new_composition
NaCaF3	NaCaI3	p-31m	I→F	9.788	0.915	high	new_composition
MgAl2F8	MgB2F8	p-3m1	B→Al	9.775	0.865	high	new_composition
LiMgF3	LiMgBr3	p-31m	Br→F	9.694	0.955	high	new_composition
NaSrF3	NaSrI3	p-31m	I→F	9.174	0.930	high	new_composition
BOF	GaOF	pmmn	Ga→B	8.913	0.505	high	new_composition
Be(BH4)2	Mg(BH4)2	p-3m1	Mg→Be	8.437	0.615	medium	new_composition
ZrZnF6	HfZnF6	p312	Hf→Zr	8.253	0.685	low	new_composition
BeClF	CaClF	p4/nmm	Ca→Be	7.756	0.840	high	new_composition
NaCaCl3	NaCaI3	p-31m	I→Cl	7.668	0.825	high	new_composition
LiBeCl3	LiMgCl3	p-31m	Mg→Be	7.610	0.960	medium	new_composition
NaSrCl3	NaSrI3	p-31m	I→Cl	7.578	0.810	high	new_composition
ScClF2	ScBr2Cl	cm11	Br→F	7.470	0.605	high	new_composition
SiPO3F	GePO3F	p11a	Ge→Si	7.292	0.620	low	new_composition
LiCaCl3	LiMgCl3	p-31m	Mg→Ca	7.208	0.855	high	new_composition
Sr(BH4)2	Mg(BH4)2	p-3m1	Mg→Sr	7.190	0.565	high	new_composition

formula	proto_formula	proto_layergroup	substitution	gap_pred	s_chem_rf	substitution_risk	novelty_class
Ca(BH4)2	Mg(BH4)2	p-3m1	Mg→Ca	7.170	0.510	high	new_composition
PCO3F	GePO3F	p11a	Ge→C	7.165	0.635	high	new_composition
SiPClO3	GePClO3	p11a	Ge→Si	7.148	0.585	low	new_composition
LiClO3	RbClO3	p2_1/m11	Rb→Li	7.118	0.870	high	new_composition
PHCO3	GePHO3	p11a	Ge→C	7.062	0.720	high	new_composition

Arquivos principais

- `outputs/candidates_guided.csv`
- `outputs/candidates_known_material.csv`
- `outputs/candidates_known_composition_new_layergroup.csv`
- `outputs/candidates_new_composition.csv`
- `outputs/top_novel_candidates.csv`
- `outputs/gap_bin_samples_all.csv`
- `outputs/gap_bin_samples_new_composition.csv`
- `outputs/relaxation_results.csv`
- `outputs/external_dft_comparison.csv`

Figuras

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